

# KUO-HAN HUNG

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## Education

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### Stanford University

June 2027 (Expected)

*M.S. in Computer Science*

*Stanford, CA*

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

### National Taiwan University

Dec. 2024

*B.S. in Computer Science and Information Engineering*

*Taipei, Taiwan*

- GPA: 4.24/4.3. **Awards:** Bachelor Thesis Dean's Award, Presidential Award (×2), Dean's List (×3).

## Technical Skills

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- **Robotics & Infrastructure:** ROS, Linux, Docker, Git, Azure, Distributed Training.
- **Languages & Frameworks:** Python, C/C++, Golang, JavaScript, SQL, Bash, PyTorch, JAX, NumPy.
- **Research Areas:** Reinforcement Learning, Vision-Language-Action Models, LLMs & VLMs, Diffusion.

## Research Experience

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### Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

*Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn*

*Stanford, CA*

- **RL finetuning of real-time VLA policies** under inference latency. Decoupled slow VLA proposals from a fast edit policy reacting to the latest observation, raising success from 42% to 97% on four dynamic tasks. [1]
- **Sample-efficient RL finetuning of pretrained VLAs** in the real world. Built a stable system with residual action edits and Q-guided sampling, reaching 30/30 success with 19 min of online data on average. [2]
- **Scaling Q-functions** with Transformers in offline RL. Introduced attention entropy regularization to stabilize and scale Transformer value functions, gaining 43% as model size grows where prior methods fail. [3]
- **Visual representation pretraining** for reinforcement learning (ongoing).

### Robot Learning Lab

Oct. 2024 – May 2025

*Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun*

*Taipei, Taiwan*

- **Manipulation policies learned from action-free videos**, using optical flow as an action surrogate. [5]

### Communication and Multimedia Lab

May 2022 – May 2024

*Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen*

*Taipei, Taiwan*

- **Vision-instruction correlation rewards** for long-horizon manipulation: 43% over prior best. [4]
- **Behavior error detection** in few-shot imitation policies via representation patterns.

## Work Experience

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### IBM Research

June 2024 – Sept. 2024

*Research Intern, Host: Dr. Pin-Yu Chen*

*Yorktown Heights, NY*

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents**.
- Traced the effect to specific attention heads using mechanistic interpretability.
- Designed an attention-based detector, a 31% AUROC improvement over the prior state of the art.

### Microsoft

July 2023 – June 2024

*Applied Scientist Intern, Host: Dr. Chiquan Zhang*

*Mountain View, CA (Remote)*

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra**.
- Developed an LLM-powered search system that adapts to user intent and geographic context.

## Selected Publications

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(\* indicates equal contribution)

- [1] **Reinforcement Learning for Real-Time Vision-Language-Action Policies**  
Perry Dong\*, **Kuo-Han Hung\***, Dorsa Sadigh, and Chelsea Finn  
Under Review
- [2] **EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models**  
Perry Dong\*, **Kuo-Han Hung\***, Tian Gao, Dorsa Sadigh, and Chelsea Finn  
Conference on Robot Learning (CoRL), 2026
- [3] **TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse**  
Perry Dong\*, **Kuo-Han Hung\***, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn  
International Conference on Machine Learning (ICML), 2026
- [4] **VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation**  
**Kuo-Han Hung\***, Pang-Chi Lo\*, Jia-Fong Yeh\*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu  
International Conference on Learning Representations (ICLR), 2025
- [5] **Learning Skill Abstraction from Action-Free Videos via Optical Flow**  
Hung-Chieh Fang\*, **Kuo-Han Hung\***, Chu-Rong Chen, . . . , Min-Hung Chen, and Shao-Hua Sun  
ICML Workshop on Building Physically Plausible World Models, 2025